SMART FARMER

Blockchain-Based Spare Parts Verification and Traceability System for Agricultural Supply Chains

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DECLARATION OF THE CANDIDATE

I declare that this is my own work and this report does not incorporate, without acknowledgement, any material previously submitted for a degree or diploma in any other university or institution of higher learning. To the best of my knowledge and belief, it does not contain any material previously published or written by another person except where acknowledgement is made in the text.

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ABSTRACT

The increasing reliance on agricultural and mechanical equipment has amplified the importance of ensuring the authenticity, traceability, and lifecycle management of spare parts used in such systems. In many developing contexts, including Sri Lanka, the circulation of counterfeit or unverified spare parts leads to reduced machinery performance, increased maintenance costs, and significant operational risks. The Smart Farmer AI Integrated Spare Part Lifecycle Forecasting System addresses this challenge through a multi-module architecture, within which this individual component focuses on the design and implementation of a Blockchain-Based Spare Part Verification and Traceability System. This component functions as the trust and integrity layer of the overall platform, ensuring that every spare part is uniquely identifiable, verifiable, and traceable throughout its lifecycle. The system leverages a permissioned blockchain framework (Hyperledger Fabric) to maintain an immutable ledger of spare part records, including registration, ownership history, and transaction details. A hybrid architecture is adopted, combining blockchain storage for tamper-proof records with a PostgreSQL database for efficient metadata management and system performance. The implementation introduces a secure QR-based verification mechanism, where each registered spare part is associated with a dynamically generated QR code containing a cryptographically signed token. This enables real-time authenticity validation through mobile or backend interfaces, effectively preventing duplication and unauthorized replication. In addition, a two-step ownership transfer protocol is designed to ensure secure and consensual transfer of spare parts between entities, enhancing trust across the supply chain. The backend system is developed using FastAPI, integrating blockchain interactions through remote execution on a cloud-hosted Hyperledger Fabric network deployed on AWS. Experimental implementation demonstrates that the system successfully enforces data immutability, supports efficient verification workflows, and provides reliable tracking of spare part lifecycle events. Overall, this research contributes a practical and scalable solution for combating counterfeit spare parts by combining blockchain technology with modern backend architectures. The proposed system enhances transparency, security, and accountability in spare part management, while also establishing a strong foundation for integration with AI-driven predictive maintenance modules within the broader Smart Farmer ecosystem.

Keywords: Blockchain, Hyperledger Fabric, spare part verification, traceability, QR authentication, supply chain security, counterfeit prevention, lifecycle management.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
BPCAT	Blockchain-based Product-centred Confidentiality-preserving Architecture for Traceability
CI/CD	Continuous Integration / Continuous Deployment
CNN	Convolutional Neural Network
DeFi	Decentralized Finance
FYP	Final Year Project
IPFS	InterPlanetary File System
IoT	Internet of Things
LKR	Sri Lankan Rupee
MRO	Maintenance, Repair, and Overhaul
NFC	Near-Field Communication
OEM	Original Equipment Manufacturer
PII	Personally Identifiable Information
REST	Representational State Transfer
RFID	Radio-Frequency Identification
SDK	Software Development Kit
SRS	Software Requirement Specification
UAT	User Acceptance Testing
UI	User Interface
UML	Unified Modeling Language
UX	User Experience
VM	Virtual Machine
WBS	Work Breakdown Structure
ZKP	Zero-Knowledge Proof

CHAPTER 1: INTRODUCTION

1.1. Background and Motivation

Modern agricultural and industrial operations are increasingly dependent on the reliability and continuous functioning of machinery. Equipment failures caused by defective or low-quality spare parts can lead to significant downtime, financial losses, and reduced productivity. In developing regions such as Sri Lanka, where agriculture remains a critical economic sector, these challenges are even more pronounced due to limited access to reliable spare part supply chains and verification mechanisms.

The management of spare parts within such environments is often inefficient and fragmented. Farmers and machine operators frequently struggle to identify genuine components, predict demand, and maintain proper records of part usage and replacement cycles. Research indicates that irregular demand patterns and lack of structured data significantly complicate spare part management processes [2]. Additionally, the widespread availability of counterfeit components further undermines trust in the system and poses risks to both equipment performance and user safety [3], [4].

Traditional systems rely heavily on centralized databases or manual documentation, which are vulnerable to data manipulation, loss, and lack of transparency. These limitations create isolated information silos where stakeholders cannot access a unified or trustworthy view of spare part history and ownership [6]. As a result, verifying the authenticity or lifecycle of a component becomes a complex and unreliable task.

Recent technological advancements, particularly in blockchain, provide a promising solution to these issues. Blockchain technology enables the creation of decentralized, immutable, and transparent ledgers that can securely record transactions and asset histories. Studies have demonstrated the effectiveness of blockchain in ensuring traceability and preventing data tampering in various industries, including manufacturing and agriculture [1], [3]. By leveraging these capabilities, it is possible to establish a trusted digital framework for managing spare parts across their entire lifecycle.

This research is part of the Smart Spare Parts Management System for Agricultural Machinery, which integrates multiple technologies to improve spare part identification, forecasting, and verification. The focus of this component is the development of a **blockchain-based verification and traceability system**, which ensures that each spare part has a secure digital identity, verifiable origin, and transparent lifecycle record.

1.2. Problem Statement

The spare parts supply chain in the agricultural sector suffers from a lack of transparency, security, and reliable verification mechanisms. Existing practices do not provide an effective way to confirm the authenticity or trace the lifecycle of spare parts, leading to significant operational inefficiencies and trust issues.

One of the primary challenges is the inability to verify whether a spare part is genuine or counterfeit. Current verification methods are largely manual and rely on physical inspection or documentation, both of which can be easily manipulated. This allows counterfeit components to enter the market undetected, resulting in reduced equipment reliability and increased maintenance costs [3], [5].

Another major issue is the absence of a structured system for tracking ownership and lifecycle events. Information related to part usage, maintenance, and transfer is often incomplete or scattered across different stakeholders. This fragmentation prevents the creation of a consistent and reliable history for each component, limiting the ability to make informed decisions regarding maintenance and replacement [6], [7].

Additionally, centralized systems introduce risks related to data integrity and system reliability. Since all data is controlled by a single authority, it becomes vulnerable to unauthorized changes and single points of failure. Research has shown that decentralized approaches are more effective in ensuring data security and trust among multiple stakeholders [8].

Therefore, the main problem addressed in this research is the lack of a secure, decentralized, and transparent system for managing spare part authenticity, ownership, and lifecycle information in a way that is both reliable and accessible to all stakeholders.

1.3. Research Objectives

The primary objective of this research is to design and develop a blockchain-based system that ensures secure verification and traceability of agricultural machinery spare parts.

The specific objectives are as follows:

- To design a decentralized system architecture that integrates blockchain technology with spare part management processes
- To develop a mechanism for assigning unique digital identities to spare parts and securely recording them on a blockchain network
- To implement a QR-based verification system that allows users to validate spare part authenticity in real time
- To design a secure ownership transfer mechanism that ensures controlled and verifiable transitions between stakeholders

- To incorporate lifecycle tracking features, including maintenance records and transaction history, into the blockchain system
- To evaluate the system in terms of security, transparency, and usability in real-world scenarios

1.4. Scope of the Research

The scope of this research is defined to ensure a focused and practical implementation of the proposed system.

- **System Functionality:** The study focuses on spare part verification, traceability, and ownership management using blockchain technology
- **Technology Implementation:** The system utilizes Hyperledger Fabric as the blockchain framework, along with backend services for system integration
- **Verification Approach:** QR code-based identification is used as the primary method for linking physical spare parts to their digital records
- **Data Management:** A hybrid approach is adopted, where critical transaction data is stored on-chain, while large or non-essential data is maintained off-chain to improve efficiency [3]
- **Application Context:** The system is designed for agricultural machinery spare parts but can be extended to other industries with similar requirements
- **Deployment Environment:** The solution is implemented in a cloud-based environment with support for integration into mobile applications

1.5. Justification and Significance

This research holds significant value in both practical and technological contexts.

From a practical perspective, the proposed system directly addresses the issue of counterfeit spare parts by enabling secure and transparent verification. By ensuring that each component can be traced and validated, the system improves trust among stakeholders and reduces the risks associated with unreliable parts. This can lead to improved machinery performance, reduced maintenance costs, and increased operational efficiency.

From a technological standpoint, the research demonstrates the application of blockchain in solving real-world supply chain challenges. Previous studies have highlighted the effectiveness of blockchain in ensuring traceability and data integrity across industries such as manufacturing, aviation, and energy systems [7], [9]. This research extends those concepts into the agricultural domain, where such solutions are still limited.

Furthermore, the integration of smart contracts, decentralized storage, and QR-based verification creates a comprehensive system that balances security, efficiency, and usability. The use of a hybrid architecture, combining on-chain and off-chain data storage, aligns with established best practices for scalable blockchain systems [3].

The broader significance of this research lies in its contribution to the digital transformation of agriculture. By introducing a transparent and secure spare part management system, it supports the development of more reliable and efficient farming operations. Additionally, the proposed approach provides a scalable framework that can be adapted to other sectors facing similar challenges in supply chain management.

CHAPTER 2: METHODOLOGY

2.1. Overview of the Research Methodology

The methodology adopted in this research was structured as a comprehensive, multi-stage system design and implementation process, supported by qualitative field investigation and iterative system refinement. The overarching approach was solution-driven and grounded in real-world applicability: by analysing existing spare part supply chain practices and identifying their limitations, the research aimed to design and implement a blockchain-based framework capable of ensuring secure verification, ownership traceability, and lifecycle transparency of spare parts.

Unlike purely theoretical or simulation-based approaches, this research integrates both qualitative and technical methodologies. A preliminary field study was conducted to understand how spare parts are currently purchased, verified, and exchanged within agricultural and mechanical environments. This decision was motivated by the understanding that supply chain challenges, particularly in developing regions, are heavily influenced by informal practices, trust-based interactions, and lack of structured data systems.

Existing literature highlights that traditional spare part management systems are highly fragmented, lack transparency, and are vulnerable to counterfeit products due to centralized control and weak verification mechanisms [3], [4]. Furthermore, the absence of standardized lifecycle tracking systems results in inefficient maintenance planning and poor decision-making [6], [7]. However, these studies often fail to capture the practical realities of how transactions occur at the user level, especially in local agricultural markets.

To address this gap, direct interactions were conducted with farmers, spare part sellers, and buyers. These interactions revealed that spare part transactions are predominantly informal, with minimal documentation and no reliable verification mechanisms. Farmers rely heavily on trust and prior experience, while sellers operate without standardized systems for tracking ownership or maintaining transaction history.

The insights gathered from this field study played a critical role in shaping the proposed system architecture. In particular, they influenced the design of a QR-based verification mechanism for ease of use, as well as the development of a controlled ownership transfer workflow that reflects real-world buying and selling interactions. The need for a tamper-proof and decentralized record system further justified the adoption of blockchain technology, which has been widely recognized for improving traceability and data integrity in supply chain applications [1], [3].

Following the requirement analysis phase, the research proceeded with the design and implementation of a hybrid system architecture integrating blockchain and off-chain storage. Hyperledger Fabric was selected as the blockchain framework due to its permissioned architecture, enabling controlled participation while ensuring data security and privacy [8]. Smart contracts were developed to automate key processes such as spare part registration, verification, and ownership transfer.

The system was implemented using a backend service developed in FastAPI, which facilitates communication between user interfaces and the blockchain network. A QR-based authentication mechanism was integrated to enable real-time verification of spare parts through secure token validation. The blockchain network and smart contracts were deployed on a cloud-based AWS EC2 environment to ensure scalability and real-world applicability.

The following broad stages constituted the full research workflow:

- Identification of spare part supply chain challenges through consultations with farmers, sellers, and buyers
- Analysis of real-world purchasing, verification, and ownership transfer practices
- Design of a blockchain-based system architecture integrating on-chain and off-chain storage
- Development and implementation of smart contracts for spare part lifecycle management
- Design and integration of a QR-based authentication mechanism for real-time verification
- Implementation of backend services to enable blockchain interaction
- Deployment of the blockchain network and smart contracts on AWS EC2 infrastructure
- Testing and evaluation of system functionality, performance, and security

Each of these stages is elaborated in detail in the following sections.

2.2. Field Study and Requirement Analysis

2.2.1 Rationale for Field Study

The inclusion of a field study in this research was essential to ensure that the proposed system is aligned with real-world operational practices. While prior studies identify counterfeit spare parts and lack of traceability as major challenges, they do not sufficiently address how these issues are experienced by end users in local environments.

Research has shown that supply chain inefficiencies often arise from fragmented information systems and lack of transparency among stakeholders [6]. Additionally, counterfeit products remain a persistent issue due to the absence of reliable verification mechanisms [3], [5]. However, these findings are often based on generalized models and do not account for informal market structures.

Therefore, a qualitative field study was conducted to gather first-hand insights from stakeholders involved in spare part usage and trading.

2.2.2 Interaction with Farmers

Farmers were engaged to understand their experiences with spare part usage, maintenance, and purchasing decisions. These discussions revealed several critical challenges:

- Spare part purchasing is primarily based on trust rather than technical verification
- Farmers frequently encounter counterfeit or low-quality components
- Equipment failures often occur due to unreliable spare parts
- There is no structured method for tracking spare part usage or replacement history

These findings highlight the need for a system that enables quick and reliable verification while maintaining a historical record of spare part lifecycle events.

2.2.3 Interaction with Sellers and Buyers

Interactions with spare part sellers and buyers provided insights into market-level processes. The following observations were made:

- Spare parts are sold without standardized authentication mechanisms
- Transactions are conducted informally, without digital documentation
- Ownership transfers are not recorded or tracked
- Buyers lack visibility into product origin and authenticity

These findings align with previous studies that identify lack of traceability and transparency as major limitations in supply chain systems [7].

2.2.4 Requirement Derivation and Design Implications

Based on field observations and literature findings, the following requirements were identified:

- A secure and tamper-proof verification mechanism
- A transparent system for tracking ownership and lifecycle events
- A user-friendly interface suitable for non-technical users
- A decentralized system architecture to ensure trust and data integrity

These requirements directly influenced the system design and technology selection.

2.3 System Design and Architecture

2.3.1 Rationale for Blockchain-Based System

Blockchain technology was selected as the foundation of the proposed system due to its ability to provide decentralization, immutability, and transparency. Unlike centralized systems, blockchain ensures that once data is recorded, it cannot be altered, thereby preventing fraud and unauthorized modifications.

Previous research has demonstrated that blockchain significantly improves traceability and reduces counterfeit risks in supply chains [1], [3], [4]. Additionally, permissioned blockchain platforms such as Hyperledger Fabric provide controlled access, making them suitable for enterprise-level applications [8].

2.3.2 Hybrid Architecture Design

The system adopts a hybrid architecture consisting of multiple layers:

- Application Layer – user interaction
- Backend Layer – FastAPI services
- Blockchain Layer – Hyperledger Fabric
- Off-Chain Storage – PostgreSQL

This approach improves system performance while maintaining data integrity [3].

2.4 Blockchain Network and Smart Contract Implementation

2.4.1 Network Setup and Deployment

The blockchain network was implemented using Hyperledger Fabric and deployed on AWS EC2. The network includes peer nodes, an orderer node, and secure communication channels. Containerization using Docker ensures scalability and flexibility.

2.4.2 Smart Contract Design

Smart contracts were developed to automate system operations:

- RegisterPart
- VerifyPart
- TransferOwnership

These contracts ensure secure and transparent transaction processing [8].

2.5 Backend Development and Integration

The backend system was implemented using FastAPI, enabling efficient communication between user applications and the blockchain network. It handles API processing, validation, and QR token generation.

2.6 QR-Based Authentication Mechanism

QR codes link physical spare parts with blockchain records. Each QR contains a secure token that ensures authenticity and prevents duplication. This approach provides a simple and effective verification mechanism suitable for real-world users [3].

2.7 Ownership Transfer Mechanism

Ownership transfer is implemented as a two-step process to ensure security and reflect real-world transaction practices. This ensures that transfers are authorized and recorded on the blockchain.

2.8 System Deployment and Integration

The system was deployed on AWS EC2 to ensure scalability and simulate real-world deployment conditions. Cloud deployment enables distributed access and supports system scalability.

2.9 System Testing and Evaluation

The system was evaluated based on:

- Functional correctness
- Performance efficiency
- Security and data integrity

Blockchain-based systems inherently provide strong guarantees of immutability and transparency, making them highly suitable for secure supply chain applications [8].

CHAPTER 3: RESULTS AND DISCUSSION

3.1 System Implementation and Functional Results

3.1.1 System Deployment and Operational Overview

The proposed Blockchain-Based Spare Parts Verification and Traceability System was successfully implemented and deployed using a combination of Hyperledger Fabric, FastAPI backend services, PostgreSQL database, and QR-based authentication mechanisms. The blockchain network was deployed on an AWS EC2 instance, ensuring real-world accessibility and scalability, while backend services were hosted in a cloud environment to facilitate API communication.

The system was evaluated through multiple functional scenarios, including spare part registration, QR-based verification, and ownership transfer processes. Each component was tested individually and as part of the integrated system workflow to ensure reliability and correctness.

The deployment environment closely simulates real-world conditions, allowing the system to be assessed not only for correctness but also for usability and performance under realistic constraints.

3.1.2 Spare Part Registration and Blockchain Recording

The first major functionality evaluated was the spare part registration process. When a new spare part is registered, the system performs two key operations:

1. Metadata storage in the PostgreSQL database
2. Transaction recording on the blockchain via smart contract invocation

The system successfully generates a unique identifier for each part and records its details on the blockchain. The blockchain transaction hash is returned and stored as a reference for future verification.

From testing, it was observed that:

- All registered parts were successfully recorded on the blockchain
- Each transaction produced a unique hash, ensuring traceability
- Data stored on-chain remained immutable across all test cases

These results confirm that the blockchain layer effectively ensures data integrity and supports the requirement for tamper-proof record keeping, as emphasized in prior research on blockchain-based supply chains [3], [4].

3.1.3 QR-Based Verification Results

The QR-based verification mechanism was evaluated by scanning generated QR codes associated with registered spare parts. Each QR code contains a secure JWT token linking the physical part to its blockchain record.

The verification workflow was tested under multiple scenarios:

- Valid QR codes
- Tampered QR tokens
- Invalid or unregistered serial numbers

The results showed that:

- Valid QR codes consistently returned correct authenticity data
- Tampered tokens were successfully detected and rejected
- Invalid serial numbers resulted in appropriate error responses

This demonstrates that the QR-based authentication mechanism provides a secure and efficient method for verifying spare part authenticity. The combination of cryptographic token validation and blockchain querying ensures both security and reliability.

3.1.4 Ownership Transfer Workflow Evaluation

The ownership transfer functionality was tested using a two-step process:

1. Buyer initiates transfer request
2. Current owner approves the request

Upon approval, the system updates both the blockchain record and the database.

The evaluation revealed:

- Ownership transfers were successfully recorded on the blockchain
- Previous ownership data remained preserved, ensuring traceability
- Unauthorized transfer attempts were prevented

This confirms that the system effectively implements a secure ownership management mechanism, addressing the lack of transaction transparency observed in traditional supply chains [6].

3.2 Performance Evaluation

3.2.1 Transaction Response Time Analysis

System performance was evaluated by measuring response times for key operations:

- Blockchain registration
- Verification queries
- Ownership transfer

It was observed that:

- Registration transactions took approximately 2–5 seconds
- Verification queries were completed within 1–2 seconds
- Ownership transfers required 3–6 seconds due to multi-step validation

These response times are acceptable for the intended use case, where real-time processing is not critical but reliability is essential.

3.2.2 System Scalability Considerations

The deployment on AWS EC2 allows the system to scale based on demand. The use of Docker containers for blockchain components further enhances scalability and flexibility.

The hybrid architecture (on-chain + off-chain storage) also contributes to performance optimization, as large datasets are not stored directly on the blockchain.

3.2.3 Reliability and System Stability

Throughout testing, the system demonstrated consistent behavior with no data loss or transaction failures. Blockchain immutability ensured that once transactions were recorded, they remained permanently accessible.

3.3 Security Analysis

3.3.1 Data Integrity and Immutability

One of the key strengths of the system is its ability to ensure data integrity through blockchain immutability. Once a spare part is registered, its data cannot be modified or deleted.

This directly addresses the issue of data manipulation found in centralized systems [3].

3.3.2 Authentication and Access Control

Security mechanisms implemented include:

- JWT-based QR authentication
- Backend validation
- Permissioned blockchain access (Hyperledger Fabric)

These measures ensure that only authorized users can interact with the system.

3.3.3 Resistance to Counterfeit and Fraud

The system effectively prevents counterfeit spare parts by:

- Linking each part to a unique blockchain record
- Enabling real-time verification
- Detecting tampered QR codes

This aligns with research highlighting blockchain as a solution for counterfeit prevention [1], [4].

3.4 Discussion

3.4.1 Effectiveness of Blockchain-Based Approach

The results demonstrate that blockchain provides a highly effective solution for spare part verification and traceability. Compared to traditional systems, the proposed approach offers:

- Enhanced transparency
- Improved trust among stakeholders
- Secure and immutable data storage

These advantages confirm the suitability of blockchain technology for supply chain applications [8].

3.4.2 Importance of Field Study Integration

The inclusion of field study insights played a critical role in system design. Features such as QR-based verification and simplified workflows were directly influenced by real-world observations.

This highlights the importance of combining technical solutions with practical user requirements.

3.4.3 System Limitations

Despite its effectiveness, the system has some limitations:

- Dependence on internet connectivity
- Limited testing scope (local deployment environment)

- Lack of large-scale user testing

These limitations provide directions for future improvements.

3.4.4 Practical Impact and Usability

The system demonstrates strong potential for real-world adoption, particularly in agricultural environments where counterfeit spare parts are a major issue.

The QR-based interface ensures ease of use, even for non-technical users, making the system accessible and practical.

CHAPTER 4: CONCLUSION AND FUTURE WORK

4.1. Summary of Findings

This report has presented the design, development, and evaluation of the Blockchain-Based Spare Part Verification and Traceability Module, which functions as the trust and security layer of the Smart Spare Parts Management System. The research was motivated by the significant operational, economic, and safety challenges associated with counterfeit spare parts and lack of traceability in agricultural machinery, particularly in smallholder farming environments where maintenance decisions are often reactive and based on limited information.

The research followed a structured system design and implementation approach, integrating field-level requirement analysis with blockchain-based architecture development. Unlike traditional centralized systems, which are vulnerable to data manipulation and lack transparency, the proposed system leverages blockchain technology to provide a decentralized, immutable, and verifiable record of spare part lifecycle events.

A key contribution of this research lies in the incorporation of real-world field insights into system design. Through interactions with farmers, spare part sellers, and buyers, it was observed that current spare part transactions are predominantly informal, lacking any reliable verification or ownership tracking mechanisms. Farmers rely heavily on trust and prior experience when purchasing spare parts, while sellers operate without standardized systems for recording transactions or verifying authenticity. These findings confirmed that the absence of a structured and secure system is a major factor contributing to the widespread presence of counterfeit components and inefficient maintenance practices.

The system implementation was evaluated across multiple functional scenarios, including spare part registration, QR-based verification, and ownership transfer workflows. The results demonstrate that the blockchain-based approach effectively addresses the identified challenges.

The first major outcome of the research is the successful implementation of a **blockchain-based registration system**, where each spare part is assigned a unique digital identity and recorded on the blockchain ledger. This ensures that once a part is registered, its data remains immutable and cannot be altered or deleted. The use of smart contracts enables automated enforcement of system rules, ensuring consistency and reliability in transaction processing.

The second key finding relates to the effectiveness of the **QR-based verification mechanism**. By linking each physical spare part to a secure digital record through a QR code containing a cryptographically signed token, the system enables fast and reliable authenticity checks. Testing results showed that valid QR codes consistently returned accurate data, while tampered or invalid codes were successfully detected and rejected. This demonstrates that the integration of QR-based authentication with blockchain querying provides a practical and secure solution for real-world verification.

The third significant outcome is the implementation of a **controlled ownership transfer mechanism**. Unlike traditional systems where ownership changes are undocumented, the proposed system records each transfer as a blockchain transaction. The two-step approval process ensures that ownership changes occur only with mutual consent, thereby preventing unauthorized transfers. Additionally, the complete ownership history of each part is preserved, enabling full lifecycle traceability.

Another important contribution of this research is the adoption of a **hybrid system architecture**, combining blockchain and off-chain storage. While critical data such as ownership and verification records are stored on the blockchain, metadata is maintained in a PostgreSQL database. This approach balances performance and security, ensuring efficient data retrieval without compromising data integrity.

From a performance perspective, the system demonstrated acceptable response times for all major operations. Blockchain transactions, including registration and ownership transfer, were completed within a few seconds, while verification operations were performed almost instantly. These results indicate that the system is suitable for real-world deployment, where reliability is more critical than ultra-low latency.

Security analysis further confirms the effectiveness of the proposed approach. The use of blockchain ensures data immutability, while JWT-based QR authentication prevents unauthorized access and duplication. The permissioned nature of Hyperledger Fabric provides controlled access, ensuring that only authorized participants can interact with the system.

Overall, the research demonstrates that a blockchain-based approach is not only technically feasible but also highly effective in addressing the challenges of spare part verification and traceability. The integration of real-world field insights with advanced technologies results in a system that is both practical and scalable.

4.2 Future Work

4.2.1 Expansion to Additional Spare Part Categories

The current implementation focuses on a generalized spare part management framework; however, future work can extend the system to support specific categories of spare parts with customized attributes and lifecycle models. For example:

- Engine components (filters, pistons, belts)
- Brake systems (pads, drums)
- Hydraulic components (hoses, valves)
- Transmission parts (chains, gears)

Each category may require specialized metadata, verification criteria, and lifecycle tracking models. Expanding the system to cover these categories would significantly increase its applicability and impact across different sectors.

4.2.2 Integration with IoT-Based Monitoring Systems

A major limitation of the current system is its reliance on user-initiated actions, such as scanning QR codes. Future work can integrate the system with IoT sensors to enable automated condition monitoring.

For example:

- Vibration sensors to detect abnormal machine behavior
- Temperature sensors to monitor overheating components
- Pressure sensors for hydraulic systems

Sensor data can be used to trigger verification requests or maintenance alerts, enabling a shift from reactive to predictive maintenance.

4.2.3 Advanced Ownership and Supply Chain Analytics

While the current system provides basic ownership tracking, future enhancements can introduce advanced analytics capabilities, including:

- Supply chain flow analysis
- Identification of counterfeit distribution patterns
- Usage trend analysis for different spare parts
- Predictive demand forecasting

These features would transform the system from a verification tool into a comprehensive supply chain intelligence platform.

4.2.4 Scalability and Large-Scale Deployment

The current implementation has been tested in a controlled environment. Future work should focus on large-scale deployment involving multiple stakeholders, including manufacturers, distributors, and service providers.

This would require:

- Optimization of blockchain performance
- Implementation of load balancing mechanisms
- Enhanced network configuration for scalability

4.2.5 User Interface and Mobile Application Development

Although the backend system is fully functional, the user interface component remains limited. Future work should focus on developing a user-friendly mobile application that simplifies interaction with the system.

Features may include:

- One-click QR scanning
- Visual verification results
- Ownership transfer notifications
- Maintenance reminders

A well-designed interface is critical for adoption among non-technical users such as farmers.

4.2.6 Integration with Government and Regulatory Systems

Another potential extension is the integration of the system with regulatory authorities to enforce standards and prevent counterfeit distribution.

This could include:

- Certification of genuine spare parts
- Monitoring of supply chain activities
- Enforcement of compliance regulations

Such integration would enhance trust and ensure wider adoption of the system.

4.2.7 Continuous System Improvement through Feedback

As the system is deployed in real-world environments, user feedback can be used to improve functionality and performance. Future work can include:

- Feedback-driven feature enhancements
- System optimization based on usage patterns
- Continuous security improvements

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